LOAN ANALYTICS

Default Prediction & Investment Analytics

A data-driven approach to smarter loan investments

The Business Problem

What are we trying to solve?

01

Who will default?

Lenders need to identify high-risk borrowers before approving loans to protect their capital.

02

How much will be lost?

When a loan defaults, the loss amount varies. Predicting this helps size risk exposure accurately.

03

Where to invest $1M?

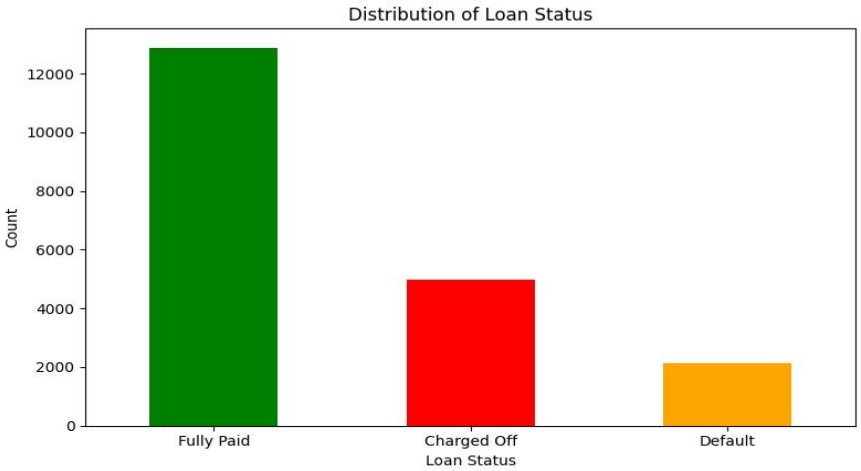
With limited funds, smart allocation across borrower segments maximises return and minimises loss.

The Dataset at a Glance

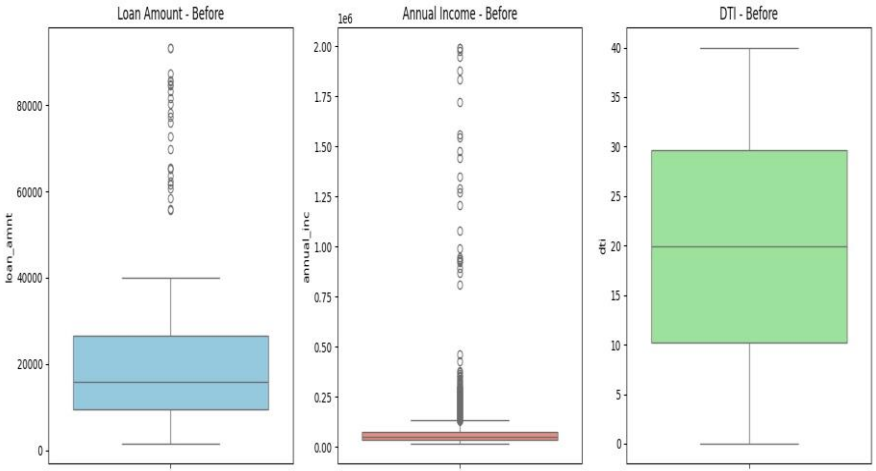
20,000 LendingClub loan records - cleaned, engineered and ready for modelling

20,000	64.4%	25.0%	10.6%
Total Loans	Fully Paid	Charged Off	Defaulted

Loan Status Distribution

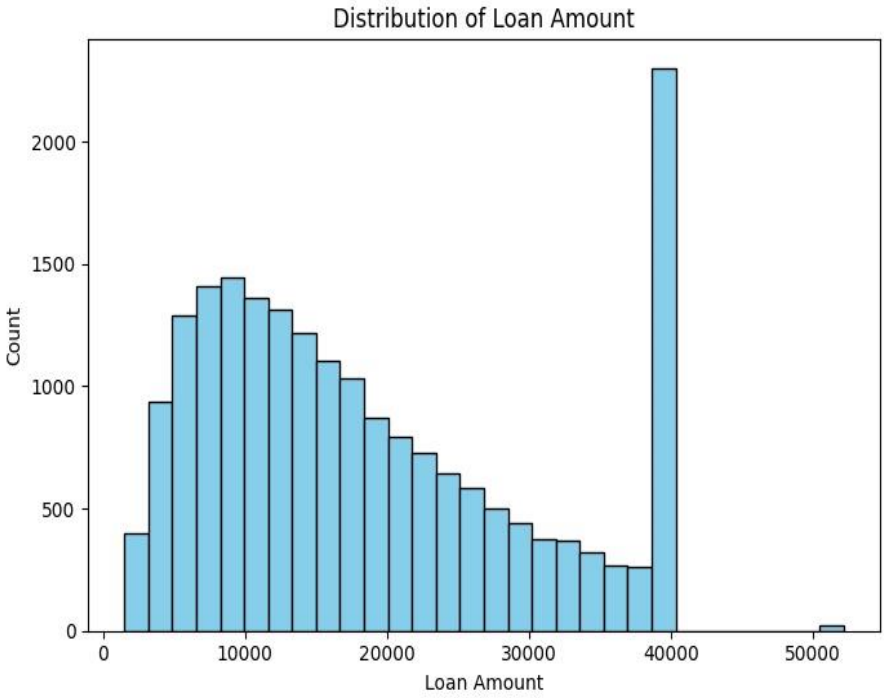


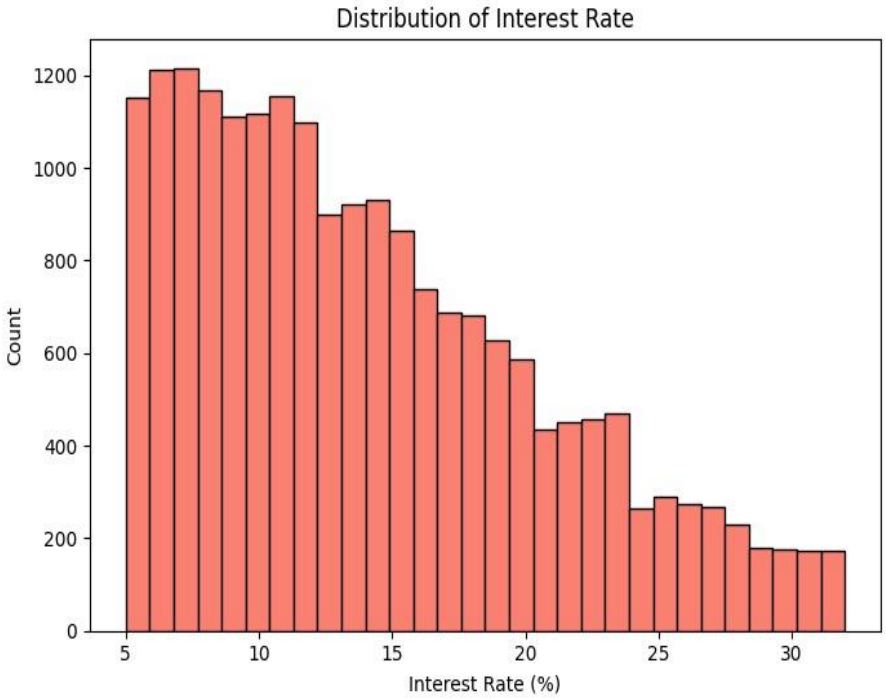
Outliers Detected Before Cleaning



Understanding the Data

How loan amounts and interest rates are distributed across borrowers

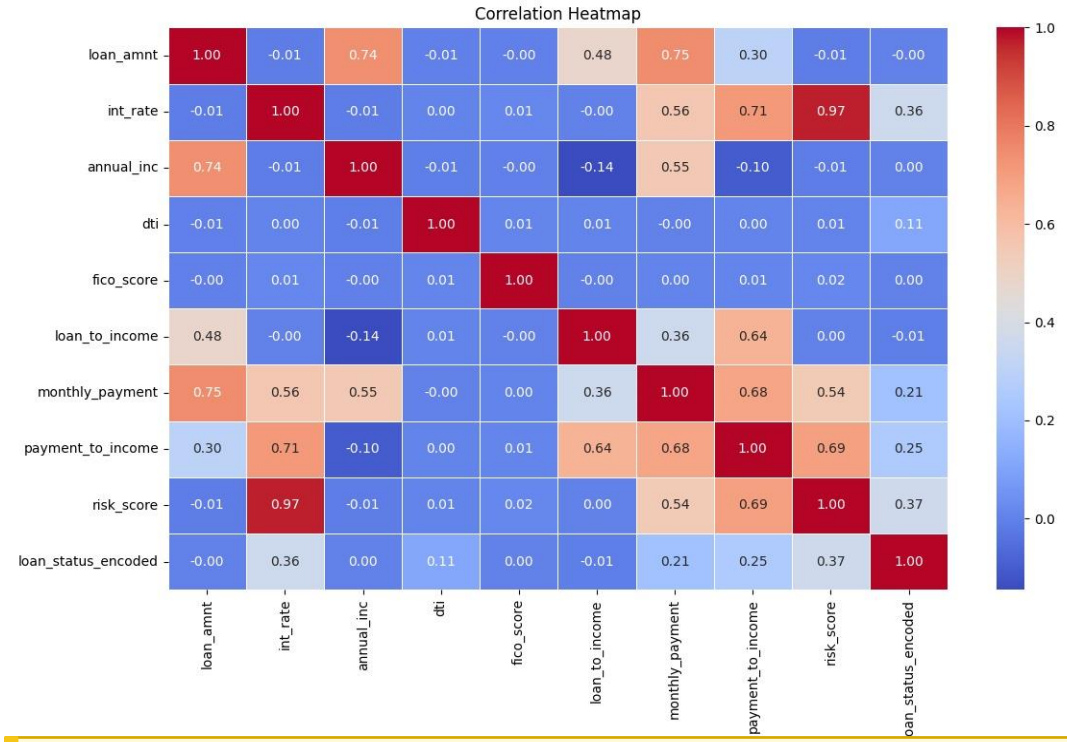




Most loans fall between $5,000 -$30,000 · Interest rates peak at 6-10% but extend to 32% for high-risk borrowers

What Drives Default?

Correlation analysis reveals the strongest predictors of loan default



0.97 **int_rate ↔ risk_score**

Near-perfect correlation — interest rate is the single strongest default signal

0.36 **int_rate ↔ default**

Higher rates are directly linked to higher default probability

0.37 **risk_score ↔ default**

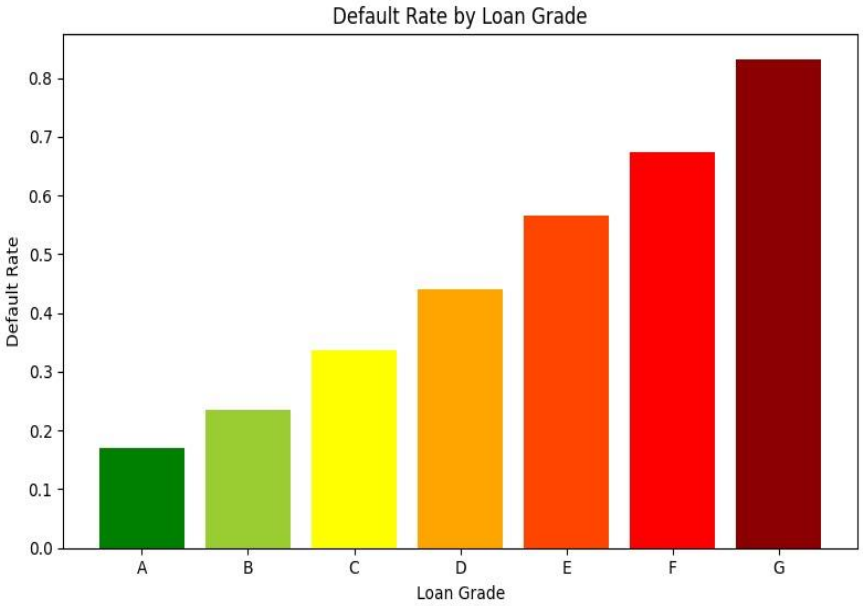
Our engineered risk_score feature is a strong predictor

Key finding: Interest rate and risk_score are the two most reliable predictors of loan default in our dataset

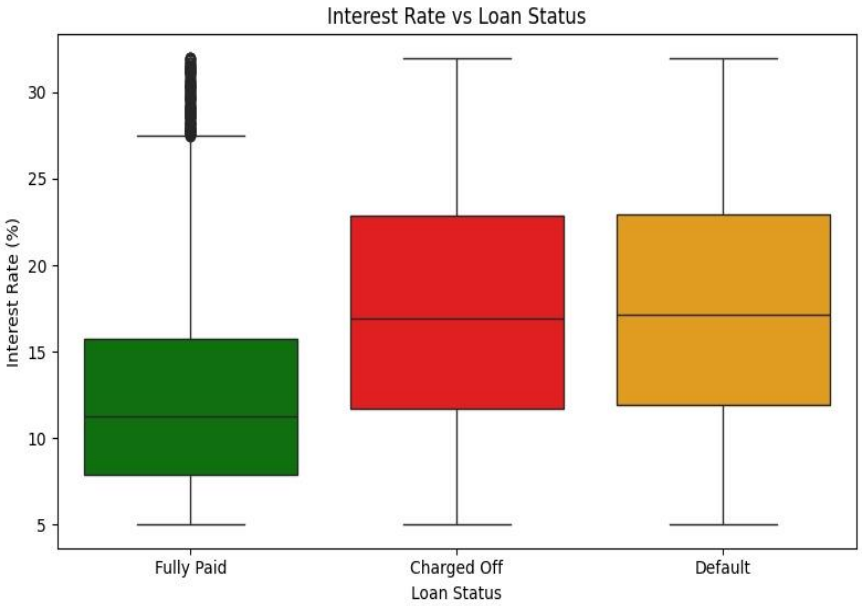
Grade & Interest Rate Tell the Full Story

Default rates climb steeply from Grade A to G — and defaulted loans carry higher interest rates

Default Rate by Loan Grade



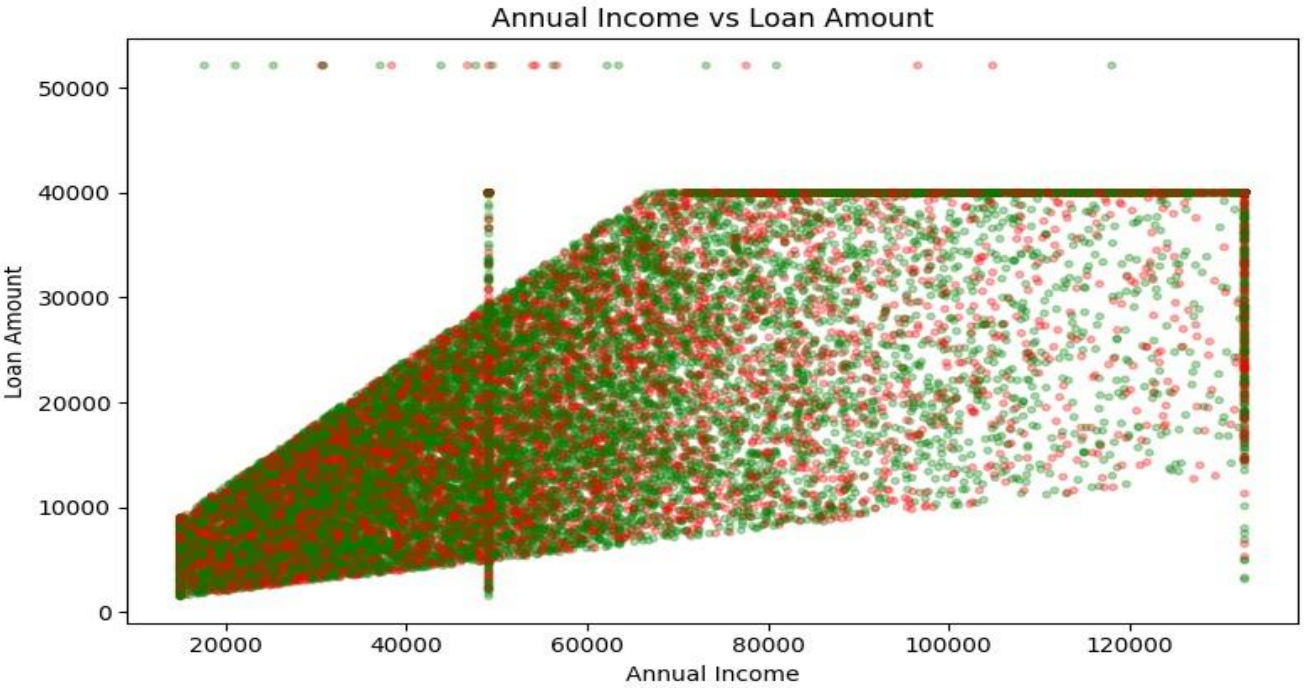
Interest Rate vs Loan Status



Grade G default rate: 83% · Grade A default rate: 17% · Defaulted loans average 17% interest vs 11% for fully paid

Income vs Loan Amount

Higher earners take larger loans — but defaults occur at all income levels

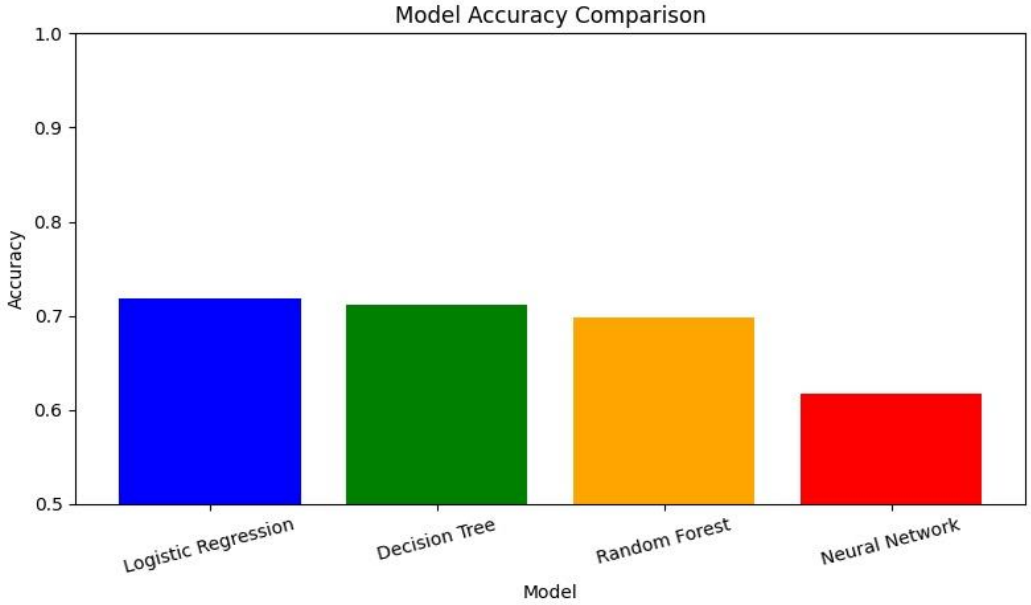


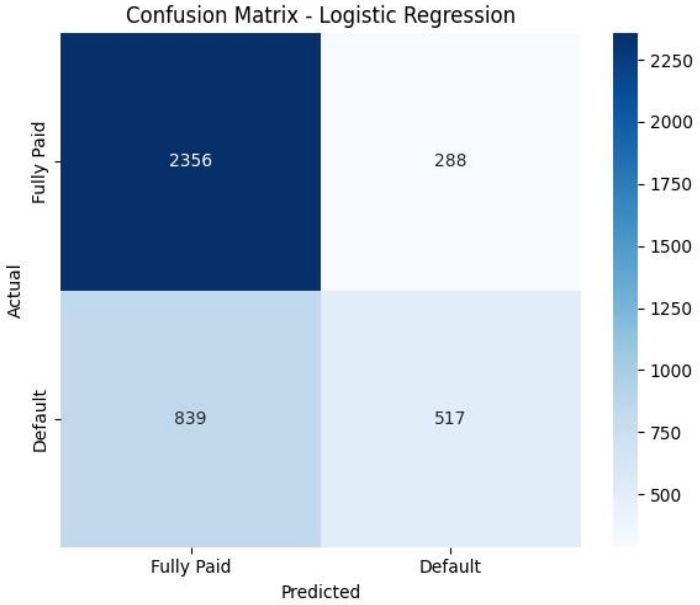
Green = Fully Paid · Red = Default · Income alone is not sufficient to predict default — rate and grade matter more

Predicting Default - Classification

We tested 4 models to predict whether a loan will default - Logistic Regression wins

Model Accuracy Comparison · 5-Fold Cross Validation · 4,000 test samples





Best Model: Logistic Regression · Accuracy 71.8% · Correctly identified 2,356 performing loans & 517 defaulting loans

Predicting Loss Amount - Regression

On defaulted loans, how much will we lose? Gradient Boosting answers this with 84.7% accuracy

Linear Regression	RMSE $3,269	R^2 0.8415
Decision Tree	RMSE $3,420	R^2 0.8266
Random Forest	RMSE $3,240	R^2 0.8443
Gradient Boosting	RMSE $3,212	**R^2 0.8470** BEST

Gradient Boosting explains 84.7% of variation in loss amounts — strong enough to size risk exposure for each loan segment

Borrower Segmentation

K-Means clustering groups 20,000 borrowers into 4 distinct risk profiles

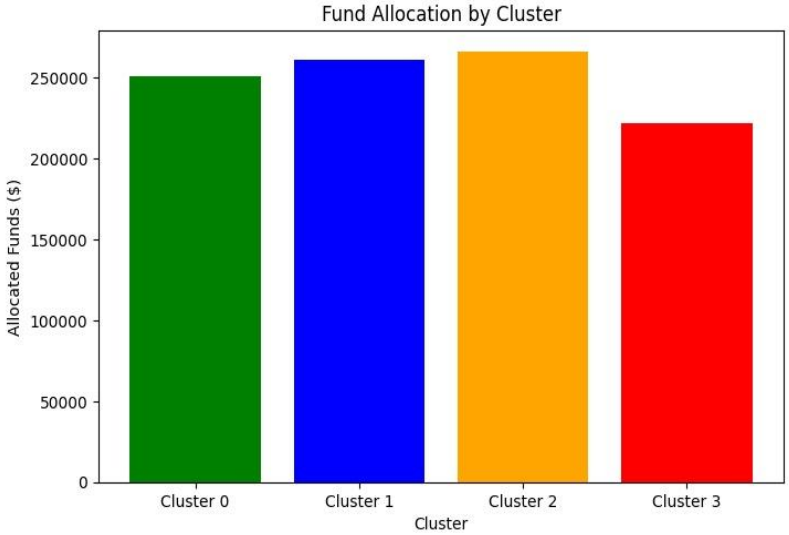
C0	5,686 loans	22%	Avg Loss $1,847	LOW RISK
C3	5,577 loans	32%	Avg Loss $2,776	MEDIUM RISK
C1	4,799 loans	34%	Avg Loss $7,748	HIGH RISK
C2	3,938 loans	62%	Avg Loss $5,846	VERY HIGH RISK

Cluster 0 (Low Risk) is the most attractive segment - highest loan count, lowest default rate, lowest average loss

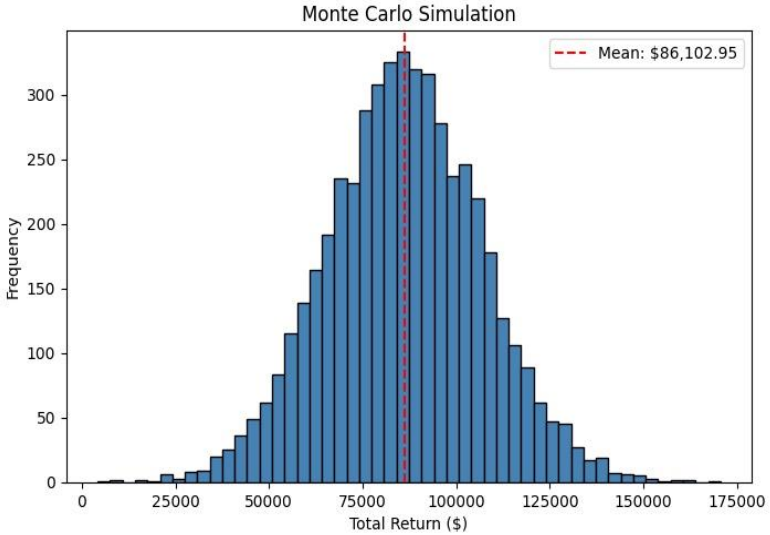
Where Should We Invest $1,000,000?

Monte Carlo simulation across 5,000 scenarios confirms our allocation strategy is robust

Fund Allocation by Cluster



Monte Carlo Simulation - 5,000 Scenarios



Mean Return	Best Case	Worst Case
$86,192	**$152,460**	**$11,888**

Recommendations

Five data-driven actions to maximise return on your loan portfolio

01

Invest primarily in Grade A & B loans

Default rates of 17-24% best risk-return balance in the portfolio

02

Screen all applicants by FICO score

Prioritise borrowers with FICO above 700 lower default probability

03

Reject Grade F & G applications outright

83% default rate on Grade G makes allocation unjustifiable at any return

04

Set DTI threshold below 20

Low debt-to-income ratio signals healthy repayment capacity

05

Concentrate 50%+ of funds in Cluster 0

Lowest default rate (22%), lowest avg loss ($1,847) proven by simulation

Thank You